

General Field Presentation Recommended Figures Pack

Sahara Kaul

Kelsey Pattison

Ahmed Sajid

CMPUT 414, Winter 2026

Contents

1	How To Use This Pack	2
2	WaveGAN: Phase Shuffle / Raw Audio GANs	2
3	MoVE: Many-to-Many Timbre Transfer Architecture	4
4	RAVE: Fidelity / Reconstruction Tradeoff	5
5	BigVGAN: Architecture Overview	6
6	DDPM: Graphical Model / Forward-Reverse Process	7
7	Classifier-Free Guidance: Guidance Strength Examples	8
8	SDEdit: Source-Preserved Editing Process	9
9	AudioLDM: Audio Manipulation by Latent Diffusion	10
10	Best Shortlist	11

1 How To Use This Pack

This document collects paper figures that are likely to be useful in the current general field presentation. For each figure, the goal is to answer three questions:

- where it should go in the presentation,
- what idea it helps explain,
- and what the original source is.

These are not meant to be the only visuals in the slide deck. They are the paper-derived figures that are most worth borrowing because they add explanatory value.

2 WaveGAN: Phase Shuffle / Raw Audio GANs

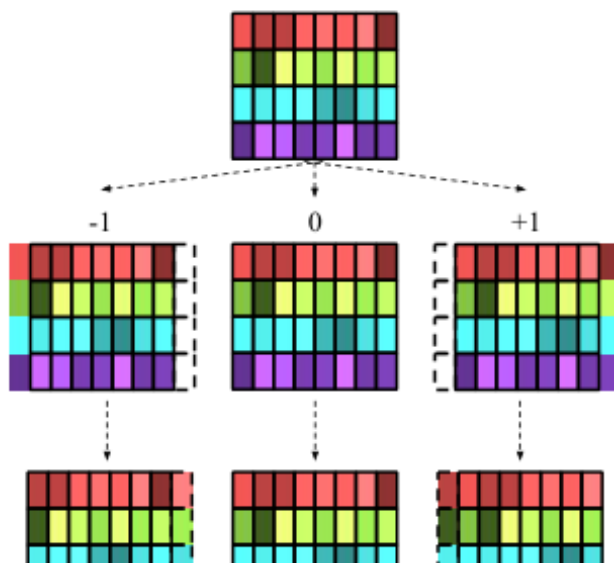
x5 2D convolution becomes length-25 1D).

itions (e.g. stride 2x2 becomes stride 4).

generator and discriminator.

et al., 2017) strategy.

ansposed
produce
s (Odena
on in im-
a learn to
ogous ar-
y overlap
, making
However,
particular
al policy
ie overall



Use it for

The *WaveGAN* slide, especially when explaining why audio GANs needed tricks like phase shuffle.

Why it helps

This figure is visually memorable and gives a concrete picture of how the phase-shuffle idea perturbs discriminator feature maps. It makes the point that raw audio GANs were not just about copying image GANs directly; they needed audio-specific adjustments.

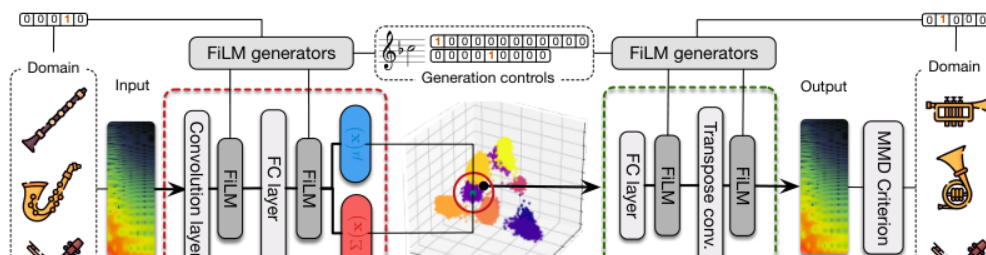
Source

Donahue, McAuley, and Puckette, *Adversarial Audio Synthesis*, ICLR 2019, Figure 3.
<https://arxiv.org/abs/1802.04208>

3 MoVE: Many-to-Many Timbre Transfer Architecture

3.2 MANY-TO-MANY TRANSFER

In order to alleviate the *one-to-one* limitation that requires a different training for each domain pair, we propose the single MoVE architecture as depicted in Figure 2. All layers are shared over the multiple domains processed, by learning a single modulated encoder E_s and decoder D_s . Transfer is performed by switching the categorical condition between different instruments. Hence, the practical success of this method highly depends on the conditioning strategy, which must also retain the pitch and octave control. To do so, we use an input embedding that jointly maps these categorical conditions to dense vectors fed into FiLM generators. We replace each intermediate batch normalization with instance normalization and activation is followed by a FiLM modulation layer (conditional instance normalization). Biasing and scaling are either applied feature-wise for 1-dimensional activations or channel-wise after 2-dimensional feature maps. A different generator output is used for modulating each instance normalization layer depending on its shape. The MoVE model trains in reconstruction with the ELBO and in transfer with the MMD, which is separately computed for each instrument against each of the others.



Use it for

The *MoVE* slide in the latent structured audio section.

Why it helps

This is one of the most useful architecture figures in the whole talk because it shows, in one place, why MoVE is more transfer-specific than a generic VAE. You can literally point to the FiLM generators and the many-to-many domain setup while talking.

Source

Bitton, Esling, and Chemla-Romeu-Santos, *Modulated Variational auto-Encoders for Many-to-Many Musical Timbre Transfer*, Figure 2.

<https://arxiv.org/abs/1810.00222>

4 RAVE: Fidelity / Reconstruction Tradeoff

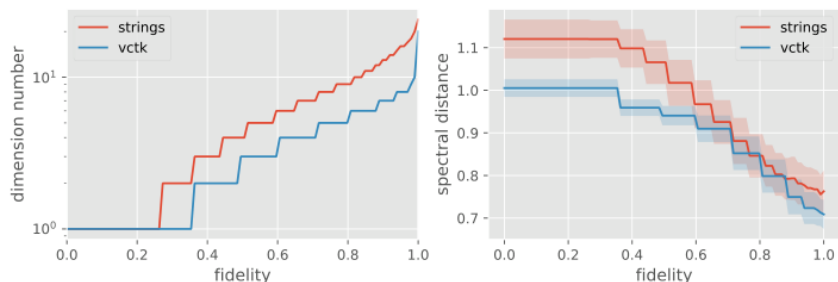


Figure 2: Estimated latent space dimensionality according to the fidelity parameter and its corresponding influence on the reconstruction quality.

size. To exemplify this, we reconstruct a sample from the *vctk* dataset using different fidelity values, and display the resulting melspectrograms in figure 3.

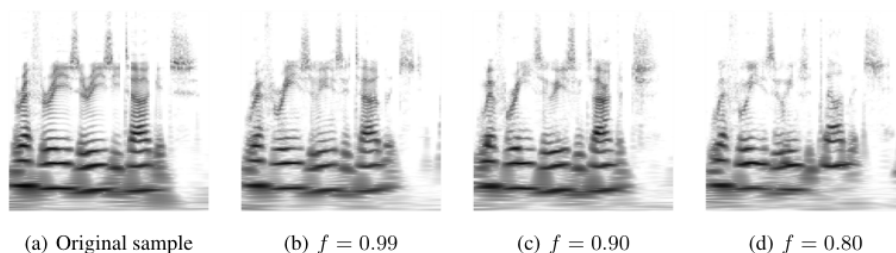


Figure 3: Reconstruction of an input sample with several fidelity parameters f .

Use it for

The *RAVE* slide, especially if you keep RAVE as a short “practical latent audio” bridge.

Why it helps

This crop shows that RAVE is really about practical latent audio compression and reconstruction quality. It is useful if you want to explain that RAVE is less about perfect disentanglement and more about making latent audio good enough to manipulate and resynthesize.

Source

Caillon and Esling, *RAVE: A variational autoencoder for fast and high-quality neural audio synthesis*, Figures 2 and 3 area.

<https://arxiv.org/abs/2111.05011>

5 BigVGAN: Architecture Overview

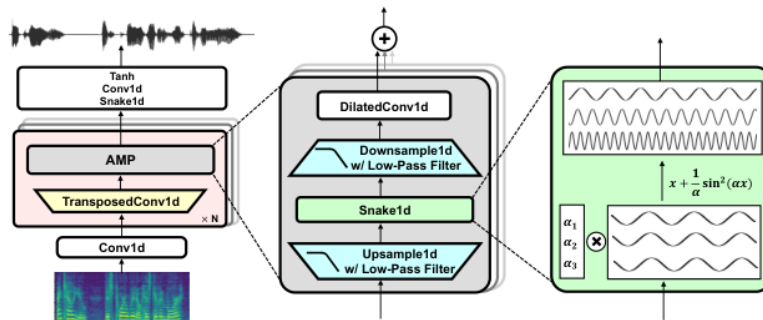


Figure 1: Schematic diagram of BigVGAN generator. The generator is composed of multiple blocks of transposed 1-D convolution followed by the proposed *anti-aliased multi-periodicity composition* (AMP) module. The AMP module adds features from multiple residual blocks with different channel-wise periodicities before dilated 1-D convolutions. It uses *Snake* function for providing periodic inductive bias, and low-pass filter for anti-aliasing purpose.

architectures can perform equally well for single-speaker neural vocoding. We demonstrate that improving generator architecture is crucial for universal neural vocoding in challenging conditions.

Use it for

The *BigVGAN* slide in the modern controllable synthesis section.

Why it helps

This figure gives a clean visual explanation of the AMP module, the Snake activation, and the audio-specific inductive bias that makes BigVGAN more than just “another vocoder.” If you are only going to show one BigVGAN visual, this is the right one.

Source

Lee et al., *BigVGAN: A Universal Neural Vocoder with Large-Scale Training*, Figure 1.

<https://arxiv.org/abs/2206.04658>

6 DDPM: Graphical Model / Forward-Reverse Process

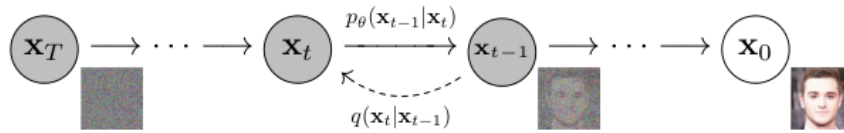


Figure 2: The directed graphical model considered in this work.

This paper presents progress in diffusion probabilistic models [53]. A diffusion probabilistic model (which we will call a “diffusion model” for brevity) is a parameterized Markov chain trained via variational inference to produce samples matching the data after finite time. Transitions of the forward process are learned to reverse a diffusion process, which is a Markov chain that gradually adds noise.

Use it for

The *DDPM* slide.

Why it helps

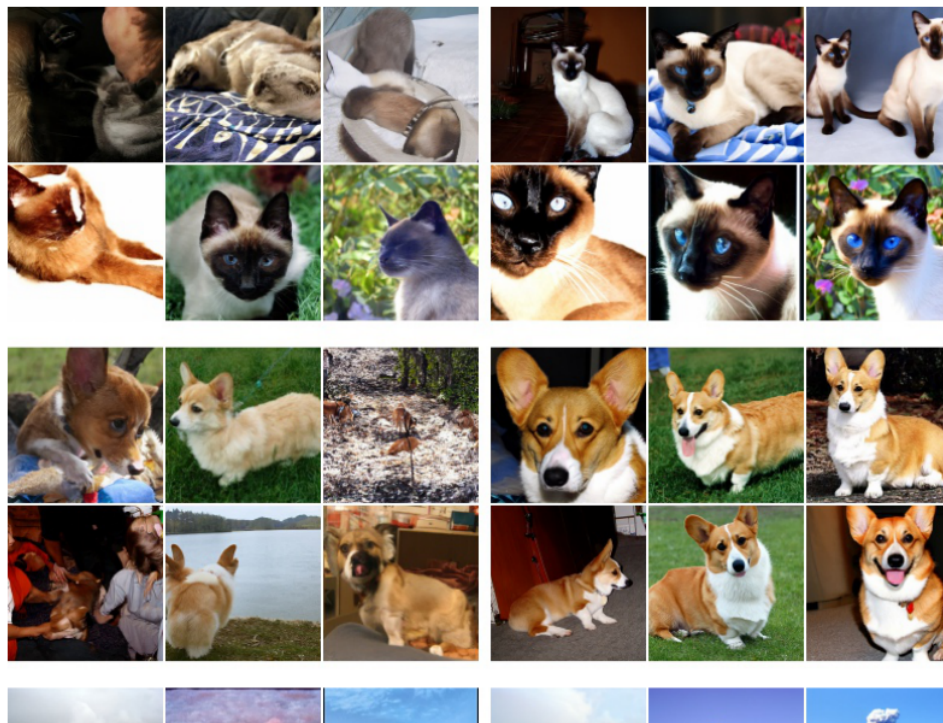
This is probably the cleanest figure to use when introducing diffusion at a high level. It gives the audience a visual model of the forward noising chain and the learned reverse denoising chain without needing to dwell too much on the math.

Source

Ho, Jain, and Abbeel, *Denoising Diffusion Probabilistic Models*, Figure 2.

<https://arxiv.org/abs/2006.11239>

7 Classifier-Free Guidance: Guidance Strength Examples



Use it for

The *Classifier-Free Guidance* slide.

Why it helps

This figure is visually strong because it lets the audience see the effect of stronger guidance immediately. Even though the domain is images, the conceptual point carries over well: larger guidance pushes harder toward the condition, sometimes at the expense of diversity or naturalness.

Source

Ho and Salimans, *Classifier-Free Diffusion Guidance*, Figure 3.

<https://arxiv.org/abs/2207.12598>

8 SDEdit: Source-Preserved Editing Process

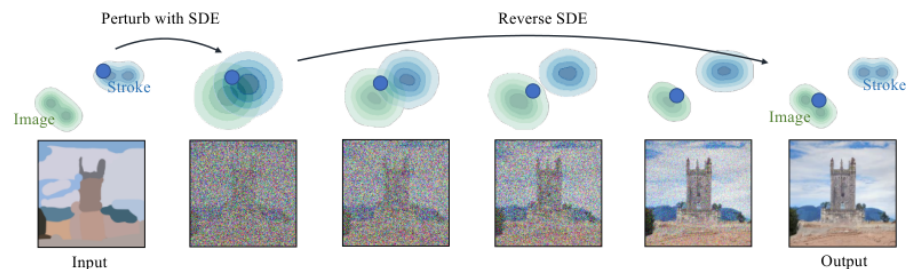


Figure 2: Synthesizing images from strokes with SDEdit. The blue dots illustrate the editing process of our method. The green and blue contour plots represent the distributions of images and stroke paintings, respectively. Given a stroke painting, we first perturb it with Gaussian noise and progressively remove the noise by simulating the reverse SDE. This process gradually projects an unrealistic stroke painting to the manifold of natural images.

Use it for

The *SDEdit* slide.

Why it helps

This is one of the best figures in the pack because it visually explains the whole idea of SDEdit in one line: perturb the source with noise, then reverse the diffusion process into a more realistic output. It directly supports your point about preserving a source skeleton while still allowing meaningful change.

Source

Meng et al., *SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations*, Figure 2.

<https://arxiv.org/abs/2108.01073>

9 AudioLDM: Audio Manipulation by Latent Diffusion

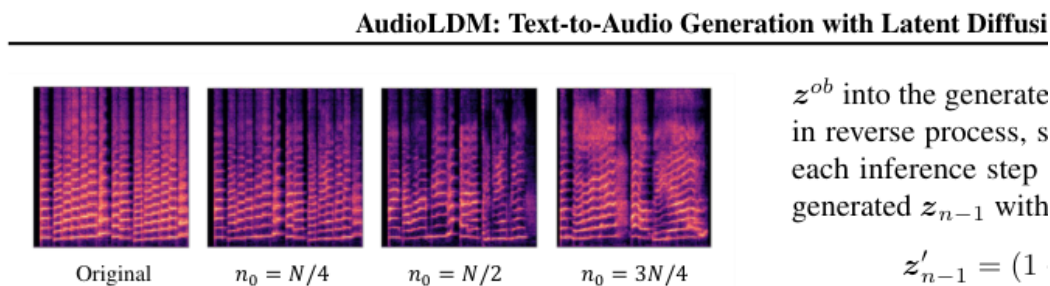


Figure 3. The manipulation result with different starting points n_0 of the shallow reverse process. The original signal is *Trumpet*, and the text prompt for style transfer is *Children Singing*.

compression level of the latent space. Our VAE is composed of an encoder and a decoder with stacked convolutional

z^{ob} into the generate in reverse process, so each inference step generated z_{n-1} with

$$z'_{n-1} = (1$$

where z' is the modified observation obtained by adding noise shown by Equation 2

The values of observation part of a mel-spectrogram

Use it for

The *AudioLDM* slide.

Why it helps

This is probably the most directly relevant modern figure for your final section. It shows that AudioLDM is not only about generation from scratch; it also supports manipulation starting from an existing audio source. That aligns very closely with your genre-remastering story.

Source

Liu et al., *AudioLDM: Text-to-Audio Generation with Latent Diffusion Models*, Figure 3.

<https://arxiv.org/abs/2301.12503>

10 Best Shortlist

If you only use a few paper figures, the most useful shortlist is:

- **MoVE architecture** for the latent-transfer section.
- **DDPM graphical model** for the first diffusion slide.
- **SDEdit overview** for the editing / source-anchoring slide.
- **AudioLDM manipulation figure** for the modern audio-diffusion slide.
- **CFG guidance examples** if you want one visually obvious control example.

If you want a slightly larger set, add:

- **WaveGAN phase-shuffle figure** for historical context.
- **BigVGAN architecture** for realism / vocoder explanation.
- **RAVE reconstruction figure** if you keep RAVE in the deck.